noc25_ee134

Mock Exam 1 | Course ID: noc25_ee134

Q1. What does the acronym EVSE stand for in the context of electric vehicle charging?

A) Electric Vehicle Safety Equipment
B) Electric Vehicle Supply Equipment
C) Electronic Voltage Supply Emitter
D) Energy Vehicle Service Enclosure

Source: -GZyYmm0B2M | Difficulty: Easy

Q2. What is the primary function of a capacitive filter when added to a diode bridge rectifier?

A) To increase the input current
B) To smooth the DC output voltage by reducing ripple
C) To convert AC to DC
D) To limit the peaky input current

Source: 1Ae4n3hWxwY | Difficulty: Easy

Q3. What is the primary motivation for operating DC-DC converters at a high switching frequency (fsw)?

A) To increase switching losses
B) To reduce the size and weight of passive components like transformers and inductors
C) To simplify the control circuit
D) To increase the output voltage ripple

Source: -xXPns3RW-g | Difficulty: Easy

Q4. In the context of EV charging, what is the primary purpose of galvanic isolation in an isolated DC-DC converter?

A) To increase the power output
B) To provide safety by electrically separating the grid from the battery
C) To reduce the size of the transformer
D) To improve the power factor

Source: TlKIeTe3Fy0 | Difficulty: Easy

Q5. What does the acronym OCPP stand for in the context of EV charging stations?

A) Open Charger Point Protocol
B) Open Charge Point Protocol
C) Online Charging Payment Protocol
D) Optimized Charge Power Protocol

Source: UAKL2GPPj0U | Difficulty: Easy

Q6. In a standard AC charging scenario, where does the primary AC-to-DC power conversion take place?

A) Within the AC EVSE
B) Within the charging cable
C) Inside the vehicle's onboard charger (OBC)
D) At a central management system (CMS)

Source: dHc3-Osl9E0 | Difficulty: Easy

Q7. In a single-phase boost PFC converter, what is the primary control variable used to regulate output voltage and achieve unity power factor?

A) Switching frequency (fsw)
B) Inductance value (L)
C) Duty ratio (d)
D) Output capacitance (C)
Source: BTGv9f8yZDI | Difficulty: Easy

Q8. What does the B-H curve for a magnetic material graphically represent?

A) The relationship between blocking voltage and holding current
B) The relationship between core loss and temperature
C) The relationship between magnetic flux density (B) and magnetic field strength (H)
D) The relationship between bandwidth and harmonic distortion
Source: CjNOVQf9HiA | Difficulty: Easy

Q9. What is the primary function of the rectifier stage in a DC EV charger?

A) To provide galvanic isolation
B) To regulate the output current
C) To convert AC power from the grid into DC power
D) To communicate with the vehicle's BMS
Source: a-fvKWBRT_g | Difficulty: Easy

Q10. What is a key advantage of the Dual Active Bridge (DAB) converter compared to the Phase-Shifted Full-Bridge (PSFB) converter?

A) It requires a larger output inductor
B) It is limited to lower power applications
C) It inherently supports bidirectional power flow
D) It has a higher risk of flux walking
Source: XsicKDSRQhk | Difficulty: Easy

Q11. What is the fundamental architectural difference between an AC EV charger and a DC EV charger?

A) AC chargers use single-phase power while DC chargers use three-phase power.
B) The power conversion from AC to DC happens onboard the vehicle for AC charging and off-board for DC charging.
C) DC chargers use CAN communication while AC chargers use PLC communication.
D) AC chargers are always low power while DC chargers are always high power.
Source: a-fvKWBRT_g | Difficulty: Medium

Q12. Why is the required output capacitance for a balanced three-phase AC-DC converter significantly smaller than for a single-phase converter of similar power?

A) The switching frequency is much higher in three-phase converters.
B) The instantaneous input power in a balanced three-phase system is constant, eliminating the second-line harmonic ripple.
C) Three-phase systems use a smaller modulation index.
D) Three-phase systems have a higher output voltage.
Source: inlEtBllhIE | Difficulty: Medium

Q13. What is the main advantage of operating a boost PFC converter in Discontinuous Conduction Mode (DCM)?

A) It achieves a lower Total Harmonic Distortion (THD) than CCM.
B) It results in lower RMS currents in the components.
C) It simplifies the control loop by eliminating the need for a multiplier block.
D) It allows for a smaller inductor size for the same power level.
Source: FBut4yhA4q0 | Difficulty: Medium

Q14. During AC Type-2 charging, how does the EV signal to the EVSE that it is ready to accept a charge?

A) By sending a high-frequency signal over the Proximity Pilot (PP) line
B) By changing the duty ratio of the PWM signal on the Control Pilot (CP) line
C) By changing the voltage level of the PWM signal on the Control Pilot (CP) line from +9V to +6V
D) By sending a CAN message to the EVSE

Source: 6-xQQwjzPIc | Difficulty: Medium

Q15. What is the primary reason for intentionally introducing an air gap into the core of a filter inductor?

A) To increase the inductance value for a given number of turns
B) To increase the core's permeability
C) To prevent core saturation by increasing its current handling capability
D) To reduce the physical size of the core

Source: 6X9uRXj0MZQ | Difficulty: Medium

Q16. What is the primary purpose of the Signal Level Attenuation Characterization (SLAC) procedure in CCS2 DC charging?

A) To measure the battery's state of charge
B) To authenticate the user for payment
C) To ensure the EV's PLC modem logically connects only with its physically connected EVSE
D) To set the maximum charging current

Source: eVsfrIYzqdg | Difficulty: Medium

Q17. What is the primary cause of Zero Crossing Distortion (ZCD) in the input current of a Totem Pole PFC converter?

A) The high switching frequency of the leading leg
B) The elimination of the input diode bridge
C) Delays and non-ideal synchronization of switches during the input voltage's zero crossing
D) The presence of a large output capacitor

Source: RxJCQteGJQA | Difficulty: Medium

Q18. In a single-phase full-bridge rectifier with a purely resistive load, what is the characteristic of the current drawn from the AC source?

A) It is a pulsating DC current.
B) It is a square wave current.
C) It is sinusoidal and in phase with the input voltage.
D) It is sinusoidal but lags the input voltage by 90 degrees.

Source: LInjt3L5-IM | Difficulty: Medium

Q19. What is a primary advantage of the JSON-based implementation of OCPP (OCPP-J) over the SOAP-based implementation (OCPP-S)?

A) It uses XML for better data structure.
B) It is more secure by default without TLS.
C) It is more scalable and efficient due to its lightweight format and use of a single WebSocket connection.
D) It requires a unique IP address for each EVSE, which simplifies networking.

Source: WUYG4Eko3rU | Difficulty: Medium

Q20. In a half-bridge converter, what is a necessary condition to achieve Zero Voltage Switching (ZVS) turn-on for a switch?

A) The load current must be zero.
B) The dead time must be long enough for parasitic capacitances to discharge the switch voltage to zero.

C) The switching frequency must be below 10 kHz.
D) The switch must be an IGBT instead of a MOSFET.

Source: -kxjMKhO51M | Difficulty: Medium

Q21. What is a major limitation of the forward converter topology that restricts its use in high-power applications?

A) It cannot provide galvanic isolation.
B) It has poor core utilization and a lossy reset mechanism that risks core saturation.
C) It can only operate in discontinuous conduction mode.
D) It requires switches with very low voltage ratings.

Source: TlKIeTe3Fy0 | Difficulty: Medium

Q22. In the DQ reference frame for three-phase converter control, what is the primary purpose of aligning the d-axis with the supply voltage space vector?

A) To maximize the current in the q-axis
B) To make the q-component of the supply voltage (Vsq) zero, simplifying control
C) To make both Vsd and Vsq non-zero
D) To eliminate the need for a PLL

Source: 3TF3ZUAokeo | Difficulty: Medium

Q23. In a three-phase AC-DC converter, the RMS current through the capacitor (`I_C_RMS`) is calculated as `√(I_converter_RMS_Line^2 - I_converter_avg_Line^2)`. What physical phenomenon does this calculation represent?

A) The total harmonic distortion of the converter current.
B) The power factor of the converter.
C) The separation of the total converter output current into its DC component (which flows to the load) and its AC component (which flows through the capacitor).
D) The energy stored in the inductor.

Source: -dgKwl8Qoqs | Difficulty: Hard

Q24. In the iterative design procedure for a high-frequency transformer, if the calculated minimum total loss (`P_total`) for a selected core is greater than the allowed total loss (`P_total_allowed`), what is the immediate next step?

A) Decrease the winding fill factor (Ku).
B) Increase the peak AC flux density (ΔB).
C) Select the next bigger core size and repeat the optimization process.
D) Decrease the number of turns in the primary winding.

Source: 24SWNw-ePds | Difficulty: Hard

Q25. In a Type-2 PWM (phase-shifted) full-bridge converter, what is the primary source of energy that enables ZVS for the *lagging* leg switches?

A) The energy stored in the large output filter inductor.
B) The energy stored in the transformer's leakage inductance.
C) The energy stored in the output capacitor.
D) The energy supplied directly from the input voltage source during dead time.

Source: XQsOeitdyiE | Difficulty: Hard

Q26. In the BEVC-DC001 standard, a 'Vehicle Statistic Data' message is identified by the PGN `001C00H`. What is the primary function of this Parameter Group Number (PGN) within the 29-bit CAN message ID?

A) To specify the destination address of the message.
B) To act as a unique identifier for the individual signal 'Final SOC'.
C) To group related parameters and allow the receiving node to quickly identify the message's context.
D) To set the priority of the message on the CAN bus.

Source: Q1IpIQJc-gw | Difficulty: Hard

Q27. During the design of a PI controller for the inner current loop of a three-phase AC-DC converter, what is the purpose of setting the controller's zero (KIC/KPC) equal to the plant's pole (Rs/L)?

A) To increase the steady-state error of the system.
B) To perform pole-zero cancellation, simplifying the system dynamics to achieve a desired bandwidth and phase margin.
C) To make the controller's response slower than the plant.
D) To ensure the controller only has a proportional response.

Source: KmgcXbLuwno | Difficulty: Hard

Q28. What is the primary reason for the large physical size of the output capacitor in a single-phase boost PFC converter operating at unity power factor?

A) The high switching frequency ripple requires a large capacitor for filtering.
B) The need to store energy for the entire line cycle during grid outages.
C) The presence of a significant low-frequency (double line frequency) ripple component in the output power.
D) The high DC output voltage requires a physically large dielectric.

Source: LGd3GukayGs | Difficulty: Hard

Q29. In the small signal model of a Phase Shifted Full Bridge (PSFB) converter, what is the significance of the 'D-loss' term?

A) It represents the conduction loss in the diodes.
B) It is a constant value that simplifies the transfer function.
C) It is a duty cycle loss term dependent on inductor current and input voltage, which complicates the control model.
D) It represents the energy lost during dead time.

Source: 568-ZYztUIY | Difficulty: Hard

Q30. In the 7-layer OSI model as applied to CCS2 High-Level Communication, which layer is responsible for assigning a logical IP address to the EV and EVSE using the Stateless Address Autoconfiguration (SLAAC) procedure?

A) Physical Layer (Layer 1)
B) Data Link Layer (Layer 2)
C) Network Layer (Layer 3)
D) Transport Layer (Layer 4)

Source: f16Qovv158l | Difficulty: Hard

Answer Key

1: B	2: B	3: B	4: B	5: B
6: C	7: C	8: C	9: C	10: C
11: B	12: B	13: C	14: C	15: C
16: C	17: C	18: C	19: C	20: B
21: B	22: B	23: C	24: C	25: B
26: C	27: B	28: C	29: C	30: C